

# Reproducible Research

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# What's (computationally) "Reproducible"?

- Someone *else* (or *you*?).
- can replicate your results given the
  - original data
  - documentation
  - any necessary code
- (but make it *easy*)
- Conflicting value: data privacy

# Our Stories

# Why Quarto (or Rmarkdown)?

- Combine code *and* text
- Generate reports, slides, websites, ebooks...
- Embed/store code without necessarily showing it
- Remember what you did
- Quickly make corrections or add new data
- ...

# What about Git/GitHub?

## What is it *for* and why might you use it?

- Store, share, and collaborate on code
- Track changes over time (who, what, when)
  - *and without my-report-final.pdf, my-report-final2.pdf...*
- Ability to revert
- Control over what to add to shared repository, when
- Know who uses and builds on your code
- Easy websites / documents hosted online
- It's *not* for: large files

# Learn More: Reproducible Research

- [A Beginner's Guide to Conducting Reproducible Research](#) (Jesse Alston & Jessica Rick)
- [Open science, reproducibility, and transparency in ecology](#) (Stephen Powers & Stephanie Hampton)
- [Best Practices for Scientific Computing](#) and [Good Enough Practices for Scientific Computing](#) (Greg Wilson & colleagues) (But seriously...`src` and `bin`?)
- [Pragmatic Reproducible Research](#) (Luke Rasmussen & colleagues)
- [Reproducibility and Replicability in Science](#) (book @ nih.gov)
- [Eleven strategies for making reproducible research...the norm](#) (Friederike Kohrs & colleagues)
- [...Data and Methods Transparency in Nonprofit Research](#) (René Bekkers & colleagues)

# Learn More: Quarto

- [Get started with Quarto](#) from quarto.org
- [Get started with Quarto](#) from us
- [Get started with Quarto](#) from *R for Data Science* (2nd ed.) by Hadley Wickham
- [Get started with Quarto in RStudio](#) from posit
- [Full Quarto documentation](#)

# Learn More: Git/GitHub

- video introducing [Git for Humans](#) (Alice Bartlett)
- [Happy git with R](#) (Jenny Bryan & colleagues)
- [GitHub Pages](#) or "[Creating a GitHub Pages Site](#)" for making a website from a repo
- [Learn Git Branching tutorials](#)

# Feedback

<https://cutt.ly/arm-feedback>

